EmoViz: Technical Report

Visualising Emotion — the complete analysis pipeline

Technical documentation of the software, experiments and findings

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This document records the technical work; the assessed dissertation is written separately by the student.

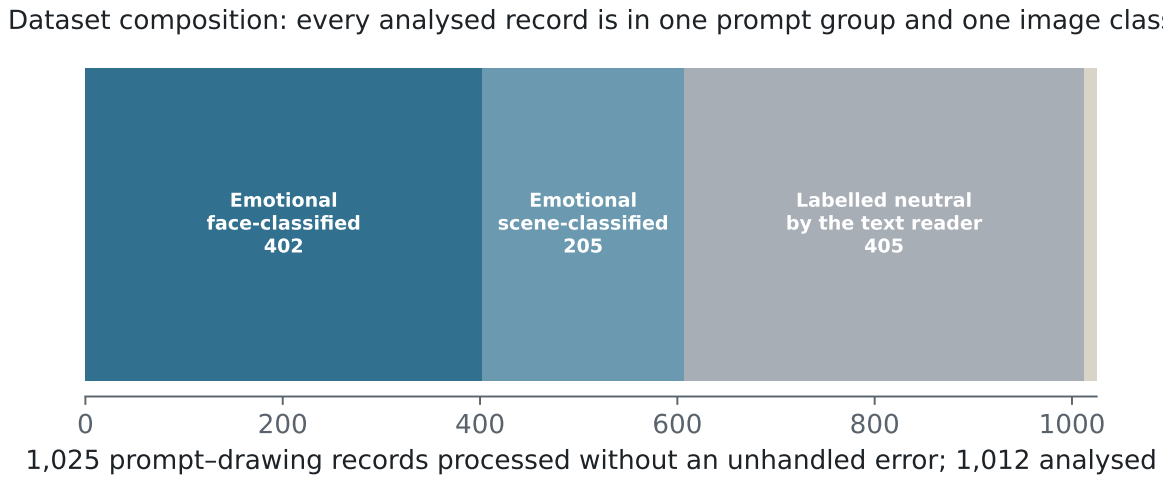
Executive summary

This project asks a question almost untouched in the text-to-image literature: **when an AI turns words about a feeling into a picture, how far does an automatic reading of the picture agree with an automatic reading of the words?** Working on a unique dataset of **1,025 prompt–drawing records** — each rendered as a black-and-white line drawing by DALL·E at a public engagement event — we built, validated and verified a complete measurement pipeline. Five findings emerged:

1. **Agreement is much higher where the describer reports a face.** Face-classified drawings agree with the prompt reader on valence sign **73.9%** of the time (family agreement 61.9%); scene-classified drawings **42.9%** (family 35.6%). Both exceed chance; the difference is +30.9 pp [+22.6, +40.6], clustered permutation $p = 0.0001$. The scene result weakens on distinct prompts.
2. **The pipeline passes its specificity check.** For prompts the text reader labels neutral (40% of the data), valence–arousal agreement sits at its null (76.6% vs 76.6%, $z = +0.3$); sign and family are degenerate there and carry no z.
3. **The same words do not reliably give the same reading.** Across 87 prompts drawn 2–42 times each, the image reader returns the same family for 68.8% of drawings within a group against a permutation null of 59.4% ($z = +4.5$); 28 of 87 groups are fully consistent against 14.5 by chance.
4. **Method matters — describe-then-extract agrees far more closely.** Like for like, the describe-then-extract reader reaches 53.0% family agreement against 22.4% for asking CLIP directly (+30.6 pp [+25.5, +35.6], cluster-level $p < 0.0001$).
5. **The reading rests on one word.** A template ablation (NLI stage only re-run) shows that removing the describer's expression sentence collapses family agreement from 53.0% to 20.3%, the other clauses move it 1–3 pp, and under a people-aware template the scene reading falls to its null (4.7% vs 3.4%, $z = +1.6$): faces, and only faces, for this reader.

All results are measured against random-pairing null baselines with clustered-bootstrap 95% intervals, re-audited on 1 and 9 September 2026 with the corrections recorded below, checked on deduplicated prompts, and covered by the automated self-test, whose numeric layer pins every headline value to the result file that produces it.

1 The dataset — and how every record is used



1,025 text–image pairs from Festival of the Mind 2024 (Sheffield): visitors typed how they felt; DALL·E drew each entry as monochrome line art. Median prompt length is 4 words; mixed feelings ("tired but happy") are common. Key data-engineering facts established:

- **Three incompatible filename schemes** and two lowercase extensions were unified by a robust loader (`emoviz/data.py`) — a naïve loader resolves under 20% of files.
- **Hidden provenance metadata** (C2PA) was recovered from inside every PNG, yielding per-image timestamps that identified records 1–13 as **pre-festival developer tests** (excluded from analysis; still processed).
- **All 1,025 images processed with no unhandled errors**; every analysed record belongs to exactly one prompt group (607 emotional / 405 labelled neutral) and one face/scene group (591/421) — the splits are groupings, not exclusions, and *all* groups are reported.
- 87 prompts repeat 2–42 times (336 records) — used both as a natural consistency experiment and for a deduplication robustness check.

2 Pipeline architecture

Following the supervisor's direction, the pipeline never asks a vision model for an emotion. Instead it *describes* then *extracts*:

Stage	Component	Output
0 · Split	GoEmotions (prompts) / BLIP-VQA cues (images)	emotional? · face?
1 · Describe	**BLIP-VQA**: expression, action, posture, scene	neutral description
2 · Extract	**Zero-shot NLI** (facts → feeling), both sides	top-3 emotions
3 · Compare	**Similarity ladder**, not exact match	4 metrics + null z

The similarity ladder scores each pair at four granularities — valence–arousal proximity, valence sign, Ekman family, and probability-weighted top-3 overlap — and every metric is referenced to a **random-pairing null** (300 shuffles): every value is read as distance above its own null.

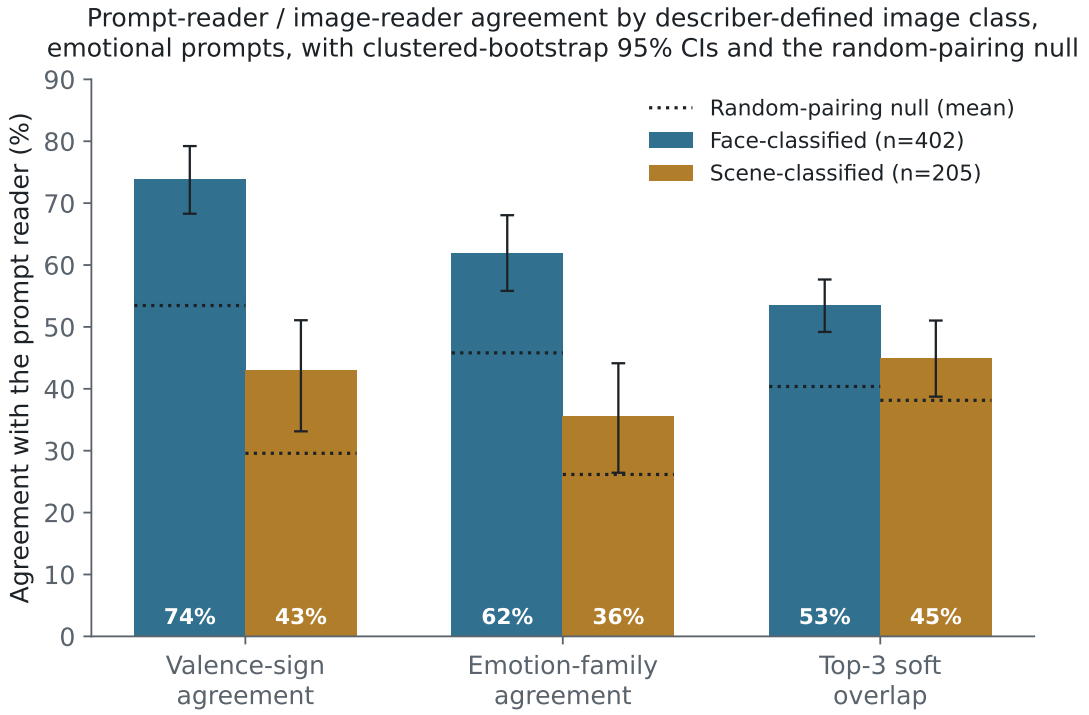
2.1 Engineering the pipeline: what was tried and rejected

Each link was tested *before* being trusted; the failures shaped the final design and are findings in their own right:

Candidate	Why rejected (evidence)
GoEmotions on image descriptions	Collapses to *neutral* 0.95+ — it needs first-person feeling words; scene descriptions have none.
Single BLIP caption → NLI	Captions too terse ("a man with hands in the air") — NLI collapses to *surprise*.
Qwen2.5-VL-3B describer	7 GB fp16 exceeded laptop RAM; thrashed into swap, zero progress in 4 h 45 min. Machine supports only sub-1B describers.
Florence-2 describer	Incompatible with transformers 5.14 (`forced_bos_token_id` failure in remote code).
BLIP-VQA (chosen)	∼0.4B, native, ∼1.5 s/record; its *"what is the facial expression?"* answer supplies exactly the affective cue NLI needs.

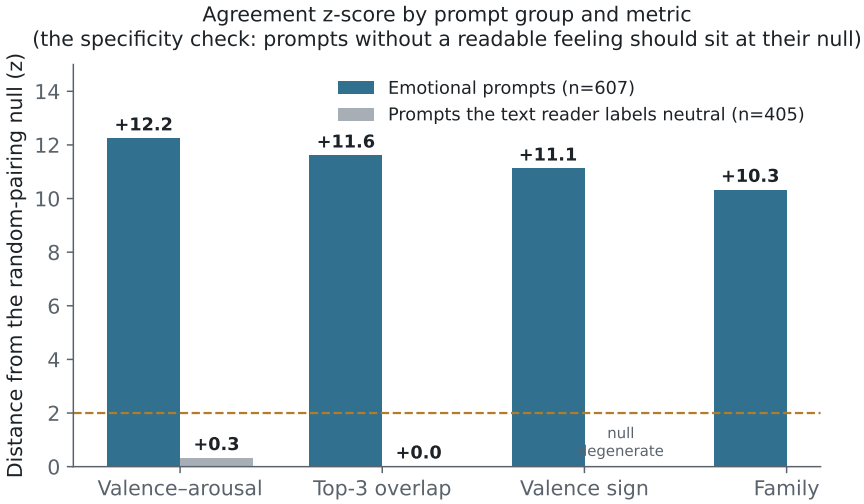
3 Findings

3.1 Finding 1 — agreement is much higher where the describer reports a face



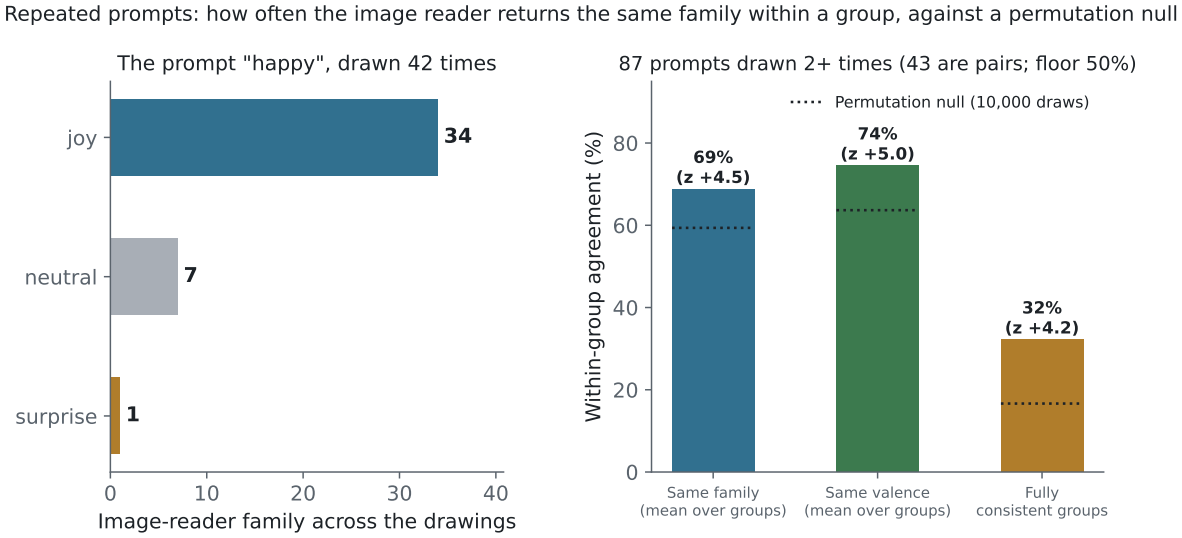
Face-classified drawings (n=402) agree with the prompt reader on valence sign 73.9% ($z = +10.4$; 95% CI [68.3, 79.2]) vs 42.9% for scene-classified (n=205, $z = +5.8$; [33.1, 51.1]); family agreement 61.9% vs 35.6%. The difference is +30.9 pp [+22.6, +40.6] (family +26.3 pp [+18.4, +35.3]), clustered permutation $p = 0.0001$. On distinct prompts the face result holds ($z = +8.4$) and the scene result weakens (valence sign $z = +3.5$, family $z = +2.6$). *Caveat (stated honestly):* the split is governed by the describer's people-count answer, and the template asserts an expression for a drawing with no person (406 of the 414 no-people scene records carry one), so the image-side reading for scenes is largely noise; the split is entangled with the measure. Finding 5 makes this concrete: under a people-aware template the scene rows fall to their null, so the contrast is faces, and only faces, for this reader.

3.2 Finding 2 — the specificity check passes



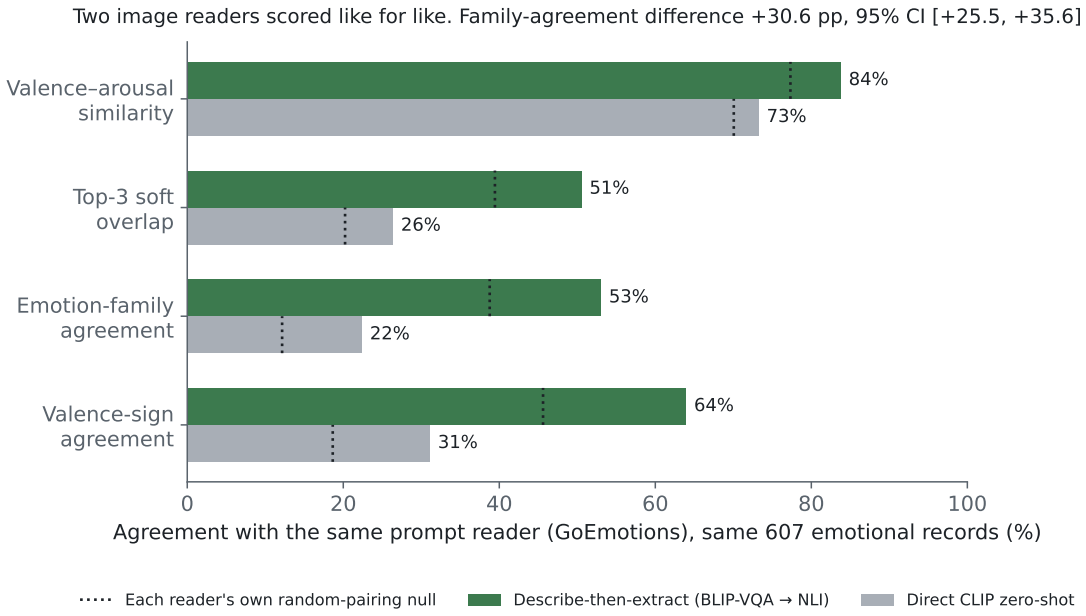
Emotional prompts sit at $z \approx +12$ on valence–arousal; prompts the text reader labels neutral ("pigs", "mickey mouse") sit at $z = +0.3$ — indistinguishable from chance. Only the valence–arousal measure is informative on that subset (sign and family are degenerate by construction); 68 of the 405 contain a plain affect word and two are a name alone, and the cleaned 335 give $z = +0.04$.

3.3 Finding 3 — the same words do not reliably give the same reading



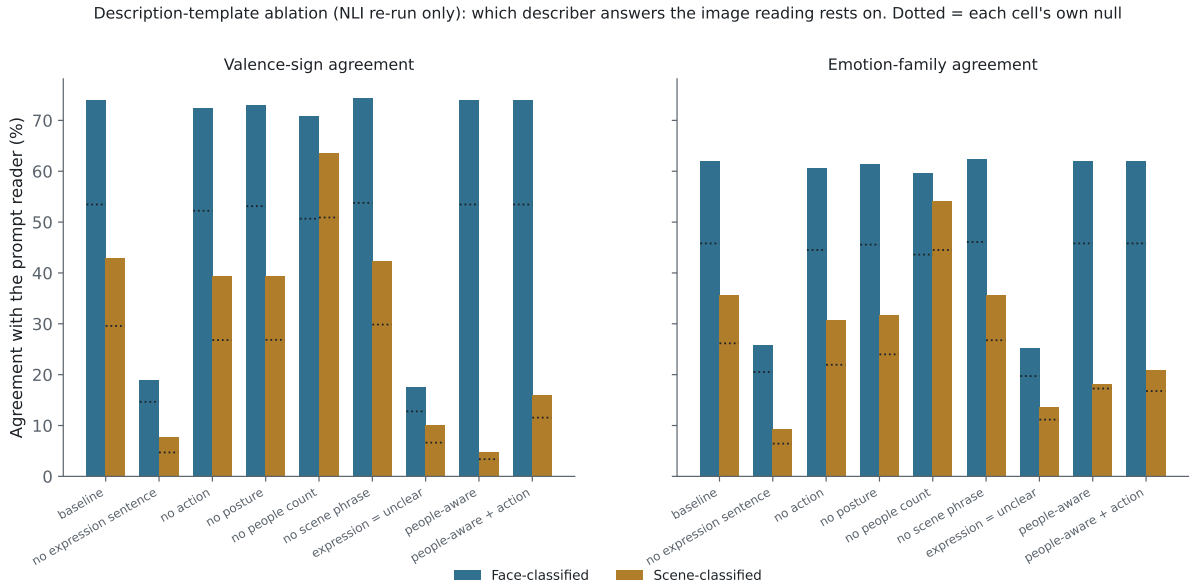
Given an *identical* prompt, the image reader returns the same family for 68.8% of drawings within a group against a permutation null of 59.4% ($z = +4.5$; valence 74.5% vs 63.7%); 28 of 87 groups are fully consistent against 14.5 by chance (these pool corpus-wide repeats — same text from different visitors and re-submissions — with generation parameters unrecorded, so the figure bounds the whole prompt→image→reader chain). Half the groups are pairs (floor 50%). "happy" ×42 produced 34 joy, 7 neutral, 1 surprise.

3.4 Finding 4 — describe-then-extract agrees far more closely



Like for like (same GoEmotions prompt reader, same 607 emotional records, same measures), describe→extract reaches 53.0% family agreement against 22.4% for direct CLIP (+30.6 pp [+25.5, +35.6], cluster-level $p < 0.0001$), 64.0% vs 31.1% on valence sign and 50.6% vs 26.4% on top-3 overlap — support for the supervisor's proposed architecture. Neither reader has ground truth: this ranks two image readers against a third model's labels. The earlier 49.3% vs 25.3% (608 records, NLI prompt side; overlap corrected from 15% to 23% after a label-space audit) is the historical v1 record.

3.5 Finding 5 — the reading rests on one word (template ablation)



Each record's five cached describer answers were re-composed under an altered template and only the NLI stage re-run (`ablate_nli.py`; a control re-run of the unmodified template reproduces every baseline number exactly). Removing the expression sentence collapses all-emotional family agreement from 53.0% to 20.3% (null 16.6%, $z = +3.8$) and valence sign from 64.0% to 15.3%; fixing the expression to "unclear" gives the same collapse; removing action, posture or scene

moves the rates by 1–3 pp; removing the people count raises rates and nulls together (scene sign 63.5% vs 50.9%, $z = +5.0$). Under a **people-aware template** ("A drawing showing {scene}." where no person is reported) the face rows are unchanged and **scene agreement falls to its null**: valence sign 4.7% vs 3.4% ($z = +1.6$), family 18.0% vs 17.3% ($z = +0.6$). The image reading rests on the describer's expression word (consistent with the 99.8% copy rate); the headline scene result was an artefact of an invented expression; the face/scene contrast is faces, and only faces, for this reader; whether scene drawings hold a feeling a person can read is what the human check tests.

3.6 What is deliberately *not* claimed

- **"Positive emotions transfer better than negative"** — retracted after re-audit: within a single-valence subset the sign/family null is degenerate ($z = 0$) and the raw gap is a base-rate artifact. The two automatic readers disagree in *opposite* directions (CLIP skews negative on line art; BLIP-VQA reports a smiling or happy expression for 57% of face-classified records and 66% of all expression answers), so only the planned **human-rating study** can adjudicate.
- Fear, disgust and surprise are recovered at very low rates by both readers, on prompt-side subsets of $n = 16$ and $n = 1$ and a surprise class largely composed of curiosity-mapped prompts; claims are restricted to overall agreement and the face/scene contrast, and no directional valence claim is made.

4 Verification and robustness

- **Null baselines everywhere.** All metrics referenced to 300 random pairings; every rate carries a clustered-bootstrap 95% interval; degenerate nulls are detected automatically and carry no z.
- **Re-audit (1 and 9 September 2026).** The analysis was re-derived twice; the corrections (degenerate nulls, missing per-metric baselines, an overstated scene claim, undisclosed duplicates, a consistency figure quoted without its null, a comparison that was not like for like, hand-typed figures) were all applied and are listed below.
- **Deduplication.** 249 duplicate-prompt records disclosed; every headline re-run on 763 distinct prompts — face finding robust, scene finding weakened.
- **Data integrity.** 1,025 unique ids, all image paths verified, all emotion fields well-formed, no unhandled errors.
- **Automated self-test** (`test.sh` + `test_numeric.py`): environment, data, redaction audit, analyses, live pipeline, pinned numeric values, figures — **PASS**.

5 Final audit

A six-dimension audit (code correctness; claim reproduction; document consistency; reproducibility; legacy-code health; deep data integrity) was run as the final step, and **every finding was fixed and re-verified**. The material findings and their resolutions:

Finding (severity)	Resolution
Label-space mismatch deflated the old method's top-3 overlap: 15% → true 23% (major)	Canonicalised `happiness→joy`; "tripled" corrected to "roughly doubled" in every document. The conclusion that describe-then-extract agrees more closely is unchanged.
Two scripts silently included the 13 pre-test records (major)	Pre-test filter added; affected table now $n = 984$ ($\leq$0.4pp; no finding changes).
README and four dashboards presented the superseded direct-CLIP era as current (critical/major)	README rewritten around the final architecture; stale dashboards and write-ups moved to `outputs/archive/` with visible supersession banners.
Face rule missed multi-word answers ("man is smiling") — ~2.5% leakage (minor)	Token-intersection matching; split refreshed (591/421), findings re-verified.
No pinned dependencies; setup script bypassed the venv; figures module missing from it (major)	`requirements.txt` (55 pinned packages); `setup.sh` now creates the venv and installs from it; figure generation added to the test suite.
Fragile old-baseline dependency (a legacy command would overwrite it) (major)	Baseline frozen as `results_clip_v2_baseline.json`; comparison script points at the frozen copy.
"Repeated-prompt runs" phrasing conflated corpus-wide repeats with consecutive runs (minor)	Reworded throughout; the pooling is now stated explicitly alongside the stricter consecutive-run count.
Assorted: two case-mismatched image paths (Linux-portability), unnormalised cached family scores, stale-cache crash path, misleading comments (minor)	All fixed; per-record paths verified against the on-disk listing; cache normalised and guarded.

The audit also *confirmed*: every published number reproduces exactly on a fresh run (deterministic, seeded); the null/z machinery is statistically sound; the data chain is airtight (1,025 unique ids, all paths resolve, zero malformed records); and the legacy baseline still runs.

Second audit (9 September 2026) against the marking criteria, with the resolutions: the consistency null was quoted from no script (now computed in `data_audit_consistency.py`, 59.4%, z+4.5); the method comparison was not like for like (re-run in `method_comparison.py` with a paired, clustered test); no confidence intervals or kappa (`inference.py`); degenerate nulls flagged by hand (now detected from the null's spread); figures typed by hand (now generated from `analysis_final.json` and its siblings); no numeric tests (`test_numeric.py`); metric code duplicated across scripts (one definition in `emoviz/metrics.py`); the control set and the describer never audited (`audits.py`); no qualitative account of disagreements (`error_taxonomy.py`); the description template never ablated (`ablate_nli.py`); no human anchor (`human_check.py`, coding pending).

6 Limitations and next steps

- **Reader bias is the binding limitation.** BLIP-VQA's positivity bias and CLIP's negativity bias bound what automatic measures can claim; the **human-rating study** is the designed adjudicator, and a 120-drawing blind single-coder check has been set up as its first step.
- Valence–arousal uses emotion-label centroids; a continuous lexicon (NRC-VAD) is a ready

upgrade.

- The face/scene rule is description-derived, and the template asserts an expression for drawings with no person; the people-aware ablation shows the scene reading sits at chance without that clause, so a scene-capable describer and a dedicated face detector are the next steps.
- Hardware bounded model choice (sub-1B describers); results are nonetheless statistically strong.

7 Repository guide

File	Role
`describe_extract.py`	The pipeline (crash-safe, resumable)
`analysis_final.py`	Final analysis: splits, similarity ladder, nulls, dedup
`data_audit_consistency.py`	Data accounting + repeated-prompt consistency
`similarity_v2.py`	Old-vs-new method comparison
`make_figures.py`	All report figures, generated from the result files
`test.sh` + `test_numeric.py`	Automated self-test: smoke layer plus pinned numeric values
`emoviz/metrics.py`	The one definition of every measure, split and null
`significance.py`, `inference.py`	Clustered test of the face/scene difference; CIs, kappa, Holm
`method_comparison.py`	Like-for-like image-reader comparison with a paired test
`audits.py`, `error_taxonomy.py`	Control and describer audits; failure-mode census
`ablate_nli.py`, `human_check.py`	Description-template ablation; blind single-coder check
`emoviz/`	Core package (loader; earlier baseline reader)
`outputs/RESULTS_final.md`	Verified findings write-up
`outputs/describe_extract.jsonl`	Full per-record pipeline output (1,025)
`outputs/analysis_final.csv`	Per-record analysis table (1,012)
`outputs/analysis_final.json`	Every reported number, machine-readable
`PIPELINE_DESIGN.md`	Architecture options & decisions

Prepared as technical documentation for the dissertation project. All numbers trace to re-runnable scripts in the repository; every figure regenerates from `make_figures.py`.